

INSTITUTO FEDERAL
SANTA CATARINA

MINISTÉRIO DA EDUCAÇÃO

SECRETARIA DE EDUCAÇÃO PROFISSIONAL E TECNOLÓGICA

INSTITUTO FEDERAL DE EDUCAÇÃO, CIÊNCIA E TECNOLOGIA DE SANTA CATARINA

LISTA DE CÁLCULO A — INTEGRAÇÃO

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ALUNO(A): _____

2 de junho de 2024

Introdução ao cálculo integral

1.) Calcule as integrais:

a.) $\int dx$.

b.) $\int x^n dx$, para $x \neq -1$.

c.) $\int \frac{1}{x} dx$.

d.) $\int (3x^2 + 6x + 1) dx$

e.) $\int \sqrt{x} dx$.

f.) $\int x^{-2/5} dx$.

g.) $\int (x^2 + 3x + 5 - 2x^{-1} + 4x^{-2}) dx$

h.) $\int_1^2 \sqrt{x} \left(x + \frac{1}{x} \right) dx$

i.) $\int \frac{5x^2 + 7x^{1/3} + 2}{x^{4/3}} dx$

j.) $\int \frac{1}{1+x^2} dx$.

k.) $\int \frac{1}{\sqrt{1+x^2}} dx$.

2.) Encontre uma função primitiva $F(x)$ da função

$f(x) = \frac{-1}{\sqrt{x^3}} + 3x^2 - 5$ que satisfaça $F(1) = 3$.

3.) Sabendo que a função $f(x)$ satisfaz a igualdade

$\int f(x) dx = \sin x - x \cdot \cos(x) - \frac{x^2}{2} + c$, determine $f(\frac{\pi}{4})$.

4.) Determine as integrais abaixo:

a.) $\int \frac{1+x^2}{x^2} dx$.

b.) $\int \frac{4x^5 - 4x^3 - 2x^2 + 1}{x^2 + 1} dx$

c.) $\int \frac{2x^5 + 14x^3 - 36x + 6}{x^2 + 9} dx$

d.) $\int 3 \sec^2 \cdot \left(\frac{\cot^2 x}{\operatorname{cosec}^2 x} + \cot^2 x \right) dx$.

e.) $\int \left(\frac{\sqrt{x} \ln(x^{10})}{x \ln(x^5)} + \frac{3x^5 + 2x^4}{x^2} \right) dx$

f.) $\int \frac{e}{\sqrt{x^3}} + 5\sqrt[5]{x^2} + \frac{\ln(x^4)}{x \ln(x)} + \pi \cdot \operatorname{cotgh}(x) dx$

g.) $\int_{-2}^2 |2x + 2| dx$

h.) $\int_{\frac{\pi}{2}}^{\pi} |\sin x| dx$

i.) $\int \sin x + 3 \cos x dx$.

j.) $\int \frac{x^2 + 1}{7} dx$.

k.) $\int \operatorname{cosec}^2 x dx$.

5.) Determine as integrais definidas abaixo:

a.) $\int_0^{\frac{\pi}{2}} \cos x dx$.

b.) $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin x + \cos x dx$.

c.) $\int_0^1 (-x^2) dx$.

d.) $\int_0^{2\pi} \sin x dx$.

6.) Calcule $\frac{dF}{dx}$, com $F(x)$ igual a:

a.) $\int_1^x (5t - 2) dt$

b.) $\int_5^x \sqrt{t} dt$

Método da substituição

5.) Calcule as integrais por substituição:

a.) $\int 3x^2 \sqrt[3]{x^3 + 1} dx$

b.) $\int (x + 1)^{17} dx$

c.) $\int e^{3x} dx$

d.) $\int \sin 7x dx$

e.) $\int x \cdot e^{x^2} dx$

f.) $\int \sec(2x) \cdot \tan(2x) dx$

g.) $\int e^{\cos x} \cdot \sin x dx$

h.) $\int \frac{1}{x+1} dx$

i.) $\int \frac{3}{(4x-7)^5} dx$

j.) $\int \frac{x}{x^2+1} dx$

k.) $\int \sin\left(5x + \frac{\pi}{2}\right) dx$

l.) $\int_4^6 \sqrt{5x-1} dx$

m.) $\int \frac{\cos \sqrt{t}}{\sqrt{t}} dt$

n.) $\int e^x \cdot \sin(e^x) dx$

o.) $\int \frac{\cos(\pi/x)}{x^2} dx$

p.) $\int \frac{x}{1+x^4} dx$

q.) $\int x \cdot \ln x dx$

r.) $\int (\cos x) \cdot (\sin^6 x) dx$

s.) $\int \tan x dx$

t.) $\int \frac{\ln x}{x} dx$

u.) $\int_0^3 x\sqrt{1+x} dx$

v.) $\int 2x \cdot \sqrt{x^2+4} dx$

w.) $\int_0^4 \sqrt{3x+4} dx$

x.) $\int 2x^2 \cdot (4-5x^3)^5 dx$

y.) $\int x \cdot \sin(x^2) dx$

z.) $\int (\sin x) \cdot \sqrt{\cos x} dx$

6.) Calcule a integral $\int \sec x dx$ por substituição, multiplicando e dividindo o integrando previamente por $\sec x + \tan x$ e usando a substituição $u = \sec x + \tan x$.

7.) A integral $\int \operatorname{cosec} x dx$ pode ser calculada de forma análoga, usando $u = \operatorname{cosec} u - \cot u$.

Integrais por partes

8.) Calcule as integrais por partes:

a.) $\int x \cdot \cos 5x dx$

b.) $\int x \cdot e^x dx$

c.) $\int_0^{\pi/3} (\cos x) \cdot (\sin 3x) dx$

d.) $\int \cos^4 x dx$

e.) $\int \sin^3 x dx$

f.) $\int \ln(x) dx$

g.) $\int x \cdot \ln(x) dx$

h.) $\int \sqrt{x} \cdot \ln(x) dx$

i.) $\int \cos \sqrt{x} dx$

j.) $\int_1^3 x \cdot e^{3x} dx$

k.) $\int x \cdot \ln(3x) dx$

l.) $\int x \cdot \cos(2x) dx$

9.) Calcule as integrais recursivas:

a.) $\int \cos(\ln(x)) dx$

b.) $\int \sec^3 x dx$

c.) $\int e^x \cdot \cos(x) dx$

10.) Sabendo que $\frac{d}{dx}(\arctg(x)) = \frac{1}{1+x^2}$, calcule a integral $\int \arctg(x) dx$ usando a técnica de inte-

gração por partes.

Para isto indica-se usar $\int 1 \cdot \arctg(x) dx$, escolhendo $u = \arctg(x)$ e $dv = 1 \cdot dx$ na fórmula

$$\int u dv = u \cdot v - \int v du$$

e por fim resolver a integral resultante por substituição.

Aplique este procedimento aos demais casos abaixo:

a.) $\int_0^{1/2} \arcsen(x) dx$

b.) $\int \arccos(x) dx$

c.) $\int \operatorname{arcsec}(x) dx$

Integrais por substituição trigonométrica

11.) Calcule as integrais por substituição trigonométrica:

a.) $\int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4-x^2}}$

b.) $\int \frac{\sqrt{x^2-25}}{x} dx$, para $x > 5$.

c.) $\int \sqrt{4-x^2} dx$

d.) $\int \sqrt{4+x^2} dx$

e.) $\int \frac{dx}{x^2 \sqrt{4x^2-9}}$

f.) $\int \frac{dx}{x^2 \sqrt{x^2+25}}$

g.) $\int \frac{x^2}{\sqrt{x^2-25}} dx$

12.) Encontre a área da elipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. Para isto considere a simetria da curva, e o fato de que a área da curva é igual a 4 vezes a sua área no primeiro quadrante.

13.) A integral $\int \frac{x}{x^2+4} dx$ pode ser calculada de duas formas, por substituição trigonométrica ou pela substituição $u = x^2+4$. Calcule-a das duas maneiras e verifique que os resultados são equivalentes.

Integrais por Frações Parciais

14.) Calcule as integrais pelo método de frações parciais.

a.) $\int \frac{2x-1}{x^2-1} dx$

b.) $\int \frac{3x-1}{(x-2)(x+5)} dx$

c.) $\int \frac{2x+4}{x^3-2x^2} dx$

d.) $\int \frac{5}{x(x^2-9)} dx$

e.) $\int \frac{dx}{x^3+x} dx$

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| <p>f.) $\int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} dx$</p> <p>g.) $\int \frac{\cos x}{(\sin x)\sqrt{1 + \sin^2 x}} dx$</p> <p>15.) Calcule a integral $\int \sec x dx$ substituindo $\sec x$ por $\frac{\cos x}{\cos^2 x}$. Utilize frações parciais e o método da substituição para obter o resultado.</p> <p>16.) Calcule as seguintes integrais pelo método do completamento de quadrado ou do algoritmo da divisão (grau do numerador maior do que o grau do denominador).</p> <p>a.) $\int \frac{x^3 - 2x}{x^2 - 1} dx$</p> | <p>b.) $\int \frac{x}{x^2 - 4x + 8} dx$</p> <p>c.) $\int \frac{1}{\sqrt{5 - 4x - 2x^2}} dx$</p> <p>d.) $\int \frac{2x^2 - 1}{(4x - 1)(x^2 + 1)} dx$</p> <p>e.) $\int \frac{x^2 - 4}{x - 1} dx$</p> <p>f.) $\int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} dx$</p> <p>g.) $\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx$</p> <p>h.) $\int \frac{x^2}{x^2 - 3x + 2} dx$</p> |
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Integrais por Completamento de Quadrado

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| <p>17.) Calcule as seguintes integrais pelo método do completamento de quadrado ou do algoritmo da divisão (grau do numerador maior do que o grau do denominador).</p> <p>a.) $\int \frac{x^3 - 2x}{x^2 - 1} dx$</p> <p>b.) $\int \frac{x}{x^2 - 4x + 8} dx$</p> <p>c.) $\int \frac{1}{\sqrt{5 - 4x - 2x^2}} dx$</p> <p>d.) $\int \frac{2x^2 - 1}{(4x - 1)(x^2 + 1)} dx$</p> <p>e.) $\int \frac{x^2 - 4}{x - 1} dx$</p> | <p>f.) $\int \frac{x^2 + x - 2}{3x^3 - x^2 + 3x - 1} dx$</p> <p>g.) $\int \frac{3x^4 + 3x^3 - 5x^2 + x - 1}{x^2 + x - 2} dx$</p> <p>h.) $\int \frac{x^2}{x^2 - 3x + 2} dx$</p> <p>i.) $\int \frac{dx}{3x^2 - 2x + 5} dx$</p> <p>18.) Calcule as integrais abaixo, usando métodos mistos, se necessário.</p> <p>a.) $\int \frac{dx}{\sqrt{x^2 + 4x}} dx$</p> <p>b.) $\int \sqrt{x^2 + 4x} dx$</p> |
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Aplicações de integrais

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| <p>1.) Esboce num mesmo sistema de eixos as funções abaixo e calcule a área da região compreendida entre as curvas:</p> <p>a.) $y = x$ e $y = x^2$.</p> <p>b.) $y = x^2 - 1$ e $y = 1 - x^2$.</p> <p>c.) $\begin{cases} x_1 = 2 \cos(t) \\ y_1 = 2 \sin(t) \end{cases}$ e $\begin{cases} x_1 = 4 \cos(t) \\ y_1 = 2 \sin(t) \end{cases}$</p> | <p>2.) Determine o volume do sólido de revolução gerado pela rotação das regiões indicadas ao redor dos eixos indicados (esboce o sólido gerado):</p> <p>a.) A região limitada pela função $y = x^2 - 3x$, pelas retas $x = -1$ e $x = 1$, girando ao redor do eixo x.</p> |
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Resoluções de alguns exercícios

Introdução ao cálculo integral

Exercício 1d

$$\int (3x^2 + 6x + 1) dx = \frac{3x^3}{3} + \frac{6x^2}{2} + x + C = x^3 + 3x^2 + x + C.$$

Exercício 1g

$$\begin{aligned} \int \left(x^2 + 3x + 5 - \frac{2}{x} + \frac{4}{x^2} \right) dx &= \\ &= \frac{x^3}{3} + \frac{3x^2}{2} + 5x - 2 \cdot \ln|x| + 4 \cdot \frac{x^{-1}}{-1} + C = \\ &= \frac{x^3}{3} + \frac{3}{2}x^2 + 5x - 2\ln|x| - \frac{4}{x} + C. \end{aligned}$$

Exercício 1h

$$\begin{aligned} \int_1^2 \sqrt{x} \cdot x + \frac{\sqrt{x}}{x} dx &= \int_1^2 x^{\frac{1}{2}} \cdot x + x^{\frac{1}{2}} \cdot x^{-1} dx = \\ &= \int_1^2 x^{\frac{3}{2}} + x^{-\frac{1}{2}} dx = \left. \frac{x^{\frac{5}{2}}}{\frac{5}{2}} + \frac{x^{\frac{1}{2}}}{\frac{1}{2}} \right|_1^2 = \\ &= \frac{2}{\frac{5}{2}} \cdot x^{\frac{5}{2}} + 2 \cdot x^{\frac{1}{2}} \Big|_1^2 = \frac{2}{\frac{5}{2}} \cdot 2^{\frac{5}{2}} + 2\sqrt{2} - \frac{2}{\frac{5}{2}} - 2 = \\ &= \frac{2}{\frac{5}{2}} \cdot 4\sqrt{2} + 2\sqrt{2} - \frac{12}{5} = \frac{18\sqrt{2}}{5} - \frac{12}{5}. \end{aligned}$$

Exercício 1i

$$\begin{aligned} \int 5x^{\frac{2}{3}} + 7x^{-1} + 2x^{-\frac{4}{3}} dx &= \\ &= \frac{5x^{\frac{5}{3}}}{\frac{5}{3}} + 7\ln|x| + 2 \cdot \frac{x^{-\frac{1}{3}}}{-\frac{1}{3}} + C = \\ &= 3x^{\frac{5}{3}} + 7\ln|x| - 6x^{-\frac{1}{3}} + C. \end{aligned}$$

Integral de substituição

Exercício 5p

$$\int \frac{x}{1+x^4} dx \quad u = x^2 \Rightarrow \frac{du}{2} = x dx$$

$$\int \frac{\frac{du}{2}}{1+u^2} = \frac{1}{2} \cdot \arctg u + C =$$

$$= \frac{1}{2} \cdot \arctg(x^2) + C$$

Exercício 5s

$$\int \operatorname{tg} x dx = \int \frac{\operatorname{sen} x}{\cos x} dx$$

$$u = \cos x \Rightarrow \frac{du}{dx} = -\operatorname{sen} x$$

$$-du = \operatorname{sen} x dx$$

$$\int \operatorname{tg} x dx = \int \frac{-du}{u} = -\ln|u| + C =$$

$$= -\ln|\cos x| + C = \ln|\sec x| + C.$$

Exercício 5v

$$\int 2x \sqrt{x^2+4} dx \quad u = x^2+4$$

$$\frac{du}{dx} = 2x \Rightarrow du = 2x dx$$

$$\int \sqrt{x^2+4} (2x dx) = \int \sqrt{u} du = \int u^{1/2} du =$$

$$= \frac{u^{3/2}}{3/2} + C = \frac{2}{3} \cdot (x^2+4)^{3/2} + C.$$

Exercício 5w

$$\int_0^4 \sqrt{3x+4} dx \quad u = 3x+4$$

$$\frac{du}{dx} = 3 \Rightarrow du = 3 dx$$

$$\int \sqrt{3x+4} dx = \int \sqrt{u} \cdot \frac{du}{3} = \frac{du}{3} = dx$$

$$= \frac{1}{3} \int u^{1/2} du = \frac{1}{3} \cdot \frac{u^{3/2}}{3/2} + C = \frac{2}{9} (3x+4)^{3/2} + C$$

$$\int_0^4 \sqrt{3x+4} dx = \frac{2}{9} (3x+4)^{3/2} \Big|_0^4 =$$

$$= \frac{2}{9} (16^{3/2} - 8) = \frac{2}{9} \cdot 56 = \boxed{\frac{112}{9}}.$$

Exercício 5x

$$\int 2x^2 (4-5x^3)^5 dx =$$

$$u = 4-5x^3$$

$$\frac{du}{dx} = -15x^2$$

$$= \int 2(4-5x^3)^5 (x^2 dx) = \frac{du}{-15} = x^2 dx$$

$$= \int 2u^5 \left(\frac{du}{-15} \right) = -\frac{2}{15} \cdot \frac{u^6}{6} + C = -\frac{1}{45} (4-5x^3)^6 + C.$$

Exercício 5y

$$\int x \cdot \operatorname{sen}(x^2) dx =$$

$$u = x^2 \therefore \frac{du}{dx} = 2x$$

$$\frac{du}{2} = x dx$$

$$= \int (\operatorname{sen} u) \frac{du}{2} = \frac{1}{2} \cdot (-\cos u) + C =$$

$$= -\frac{1}{2} \cdot \cos x^2 + C.$$

Exercício 5z

$$\begin{aligned} \int \sin x (\cos x)^{\frac{3}{2}} dx &= \\ &= \int u^{\frac{3}{2}} (-du) = -\frac{u^{\frac{5}{2}}}{\frac{5}{2}} + C = \\ &= -\frac{2}{5} \cdot (\cos x)^{\frac{5}{2}} + C. \end{aligned}$$

$$\begin{aligned} u &= \cos x \\ \frac{du}{dx} &= -\sin x \\ -du &= \sin x dx \end{aligned}$$

Exercício 6

$$\begin{aligned} \int \sec x dx &= \int (\sec x) \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx \\ u &= \sec x + \tan x \\ \frac{du}{dx} &= \sec x \cdot \tan x + \sec^2 x \\ \frac{du}{dx} &= (\sec x)(\tan x + \sec x) \\ du &= (\sec x)(\tan x + \sec x) dx \\ \int (\sec x) \frac{(\sec x + \tan x)}{(\sec x + \tan x)} dx &= \\ \int \frac{1}{u} du &= \ln|u| + C = \ln|\sec x + \tan x| + C. \end{aligned}$$

Exercício 7

$$\begin{aligned} \int \operatorname{cosec} u du &= \int \operatorname{cosec} u \left(\frac{\operatorname{cosec} u - \cot u}{\operatorname{cosec} u - \cot u} \right) du = \\ w &= \operatorname{cosec} u - \cot u \\ \frac{dw}{du} &= -\operatorname{cosec} u \cdot \cot u + \operatorname{cosec}^2 u \\ dw &= \operatorname{cosec} u (\operatorname{cosec} u - \cot u) du \\ &= \int \frac{1}{w} dw = \ln|w| + C = \ln|\operatorname{cosec} u - \cot u| + C \end{aligned}$$

Integral por partes

Exercício 8i

$$\int \cos(\sqrt{x}) dx =$$

Usando a substituição $u = \sqrt{x}$:

$$\frac{du}{dx} = \frac{1}{2\sqrt{x}} \Rightarrow \frac{du}{dx} = \frac{1}{2u} \Rightarrow dx = 2u du.$$

Voltando à integral:

$$= \int (\cos u)(2u du) = \text{Usando integração por partes:}$$

$$= 2 \left(u \cdot (\text{senu}) - \int \text{senu} du \right) = 2(u \cdot \text{senu} + \cos u) + c =$$

$$= 2(\sqrt{x} \cdot \sin \sqrt{x} + \cos \sqrt{x}) + c$$

Exercício 8j

$$\int x \cdot \underbrace{(e^{3x} dx)}_{du} = x \cdot \frac{1}{3} e^{3x} - \int \frac{e^{3x}}{3} dx =$$

$$du = e^{3x} dx \Rightarrow u = \int e^{3x} dx = \frac{1}{3} e^{3x}$$

$$v = x \Rightarrow dv = dx$$

$$= \frac{x}{3} \cdot e^{3x} - \frac{1}{9} \cdot e^{3x} + C = \frac{e^{3x}}{3} \left(x - \frac{1}{3} \right) + C.$$

$$\int_1^3 x \cdot e^{3x} dx = \frac{e^{3x}}{3} \left(x - \frac{1}{3} \right) \Big|_1^3 =$$

$$= \frac{e^9}{3} \left(3 - \frac{1}{3} \right) - \frac{e^3}{3} \left(1 - \frac{1}{3} \right) = \frac{8}{9} \cdot e^9 - \frac{2}{9} \cdot e^3.$$

Exercício 8k

$$\int x \cdot \ln(3x) dx = \int \ln(3x) \cdot (x dx) =$$

$$\boxed{dv = x dx \Rightarrow v = \frac{x^2}{2}} \quad \boxed{u = \ln 3x \Rightarrow du = \frac{3}{3x} dx}$$

$$du = \frac{1}{x} dx$$

$$= \left(\ln 3x \right) \left(\frac{x^2}{2} \right) - \int \frac{x^2}{2} \cdot \frac{1}{x} dx =$$

$$= \frac{x^2}{2} \cdot \ln 3x - \frac{x^2}{4} + C = \frac{x^2}{2} \left(\ln(3x) - \frac{1}{2} \right) + C.$$

Exercício 8l

$$\int x \cdot \cos 2x dx = x \cdot \frac{\sin 2x}{2} - \int \frac{\sin 2x}{2} dx =$$

$$\boxed{u = x \Rightarrow du = dx} \quad \boxed{dv = \cos 2x dx}$$

$$v = \frac{1}{2} \cdot \sin 2x$$

$$= \frac{x \cdot \sin 2x}{2} - \frac{1}{2} \cdot \left(\frac{-\cos 2x}{2} \right) + C =$$

$$= \frac{x \cdot \sin 2x}{2} + \frac{\cos 2x}{4} + C.$$

Exercício 9a

$$\int \cos(\ln x) dx = \int 1 \cdot \cos(\ln x) dx =$$

$$= x \cdot \cos(\ln x) - \int x \cdot \frac{1}{x} \cdot \sin(\ln x) dx =$$

$$= x \cdot \cos(\ln x) - \int \sin(\ln x) dx \quad (1)$$

$$\int \sin(\ln x) dx = \int 1 \cdot \sin(\ln x) dx =$$

$$= x \cdot \sin(\ln x) + \int x \cdot \frac{1}{x} \cdot \cos(\ln x) dx =$$

$$= x \cdot \sin(\ln x) + \int \cos(\ln x) dx \quad (2)$$

(2) em (1):

$$\int \cos(\ln x) dx = x \cdot \cos(\ln x) - \left(x \cdot \sin(\ln x) + \int \cos(\ln x) dx \right)$$

$$2 \int \cos(\ln x) dx = x(\cos(\ln x) - \sin(\ln x)) + C$$

$$\int \cos(\ln x) dx = \frac{x}{2} (\cos(\ln x) - \sin(\ln x)) + C.$$

Exercício 9b

$$\int \sec^3 x \, dx = \int \underbrace{\sec x}_u \cdot \underbrace{(\sec^2 x \, dx)}_{dv} =$$

Por partes: $u = \sec x \Rightarrow du = \sec x \tan x \, dx$
 $dv = \sec^2 x \, dx \Rightarrow v = \tan x$

$$= \sec x \cdot \tan x - \int \tan x \cdot \sec x \cdot \tan x \, dx =$$

$$= \sec x \tan x - \int \sec x \cdot \tan^2 x \, dx =$$

Relação Fundamental da Trigonometria

$$\sin^2 x + \cos^2 x = 1 \Rightarrow \tan^2 x + 1 = \sec^2 x$$

$$\tan^2 x = \sec^2 x - 1$$

$$= \sec x \tan x - \int \sec x (\sec^2 x - 1) \, dx =$$

$$= \sec x \tan x - \int \sec^3 x - \sec x \, dx =$$

$$= \sec x \tan x - \int \sec^3 x \, dx + \int \sec x \, dx =$$

De exercício anterior:

$$\int \sec x \, dx = \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x \, dx = \sec x \tan x - \int \sec^3 x \, dx + \ln |\sec x + \tan x| + C$$

$$2 \int \sec^3 x \, dx = \sec x \tan x + \ln |\sec x + \tan x| + C$$

$$\int \sec^3 x \, dx = \frac{1}{2} (\sec x \tan x + \ln |\sec x + \tan x|) + C$$

Exercício 10

$$\int (\arctan x) (1 \, dx) =$$

$$du = dx \Rightarrow u = x \quad v = \arctan x \Rightarrow dv = \frac{dx}{1+x^2}$$

$$= (\arctan x) x - \int x \cdot \frac{dx}{1+x^2} =$$

$$w = 1+x^2 \Rightarrow dw = 2x \, dx \quad \frac{dw}{2} = x \, dx$$

$$= x \cdot \arctan x - \int \frac{dw}{2w} = x \cdot \arctan x - \frac{1}{2} \cdot \ln |w| + C =$$

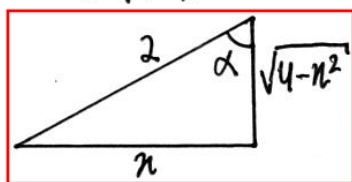
$$= x \cdot \arctan x - \frac{1}{2} \cdot \ln |1+x^2| + C =$$

$$= x \cdot \arctan x - \frac{1}{2} \cdot \ln (1+x^2) + C.$$

Integral por substituição trigonométrica

Exercício 11a

$$\int \frac{dx}{x^2 \sqrt{4-x^2}} =$$



$$\begin{aligned} \sin \alpha &= \frac{x}{2} \\ x &= 2 \sin \alpha \\ dx &= 2 \cos \alpha \, d\alpha \end{aligned}$$

$$= \int \frac{2 \cos \alpha \, d\alpha}{4 \sin^2 \alpha \sqrt{4-4 \sin^2 \alpha}} =$$

$$= \int \frac{2 \cos \alpha \, d\alpha}{(4 \sin^2 \alpha) 2 \cos \alpha} = \frac{1}{4} (-\cot \alpha) + C =$$

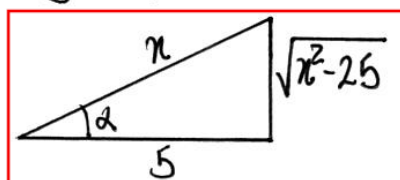
$$= -\frac{1}{4} \cdot \frac{\sqrt{4-x^2}}{x} + C$$

$$\int_1^{\sqrt{2}} \frac{dx}{x^2 \sqrt{4-x^2}} = -\frac{1}{4x} \cdot \sqrt{4-x^2} \Big|_1^{\sqrt{2}} =$$

$$= -\frac{1}{4\sqrt{2}} \sqrt{2} + \frac{1}{4} \sqrt{3} = \frac{1}{4} (\sqrt{3} - 1).$$

Exercício 11b

$$\int \frac{\sqrt{x^2-25}}{x} dx$$



$$\sec \alpha = \frac{x}{5} \Rightarrow x = 5 \sec \alpha$$

$$\frac{dx}{d\alpha} = 5 \sec \alpha \tan \alpha \Rightarrow dx = 5 \sec \alpha \tan \alpha \, d\alpha$$

$$\text{Do triângulo: } \sin \alpha = \frac{\sqrt{x^2-25}}{x}$$

$$\int \frac{\sqrt{x^2-25}}{x} dx = \int (\sin \alpha) (5 \sec \alpha \tan \alpha \, d\alpha) =$$

$$= 5 \int \tan^2 \alpha \, d\alpha \stackrel{*}{=}$$

Da relação fundamental da Trigonometria:

$$\sin^2 \alpha + \cos^2 \alpha = 1 \quad (\div \cos^2 \alpha)$$

$$\tan^2 \alpha + 1 = \sec^2 \alpha \Rightarrow \tan^2 \alpha = \sec^2 \alpha - 1$$

$$\begin{aligned} \stackrel{*}{=} 5 \int \sec^2 \alpha - 1 \, d\alpha &= 5 \left(\int \sec^2 \alpha \, d\alpha + \right. \\ &\quad \left. - \int d\alpha \right) = 5 (\tan \alpha - \alpha) + C \stackrel{**}{=} \end{aligned}$$

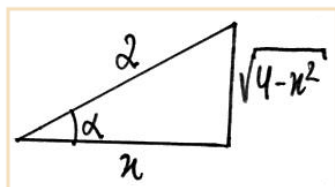
$$\text{Do triângulo: } \tan \alpha = \frac{\sqrt{x^2-25}}{5}$$

$$\sec \alpha = \frac{x}{5} \Rightarrow \alpha = \arccos \frac{x}{5}$$

$$\stackrel{**}{=} 5 \left(\frac{\sqrt{x^2-25}}{5} - \arccos \frac{x}{5} \right) + C.$$

Exercício 11c

$$\int \sqrt{4-x^2} dx$$



$$\sin \alpha = \frac{\sqrt{4-x^2}}{2}$$

$$\begin{aligned} \cos \alpha &= \frac{x}{2} \\ x &= 2 \cos \alpha \\ dx &= -2 \sin \alpha d\alpha \end{aligned}$$

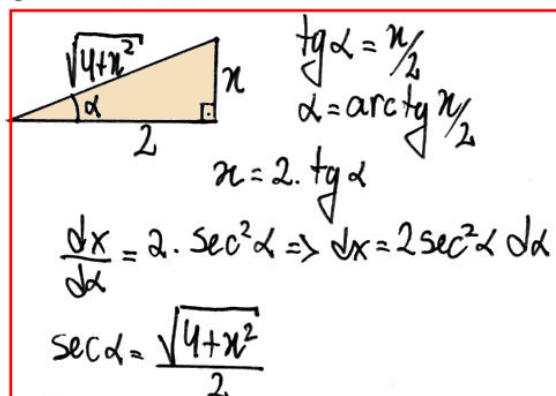
$$\begin{aligned} \int \sqrt{4-x^2} dx &= \int 2 \sin \alpha (-2 \sin \alpha d\alpha) = \\ &= -4 \int \sin^2 \alpha d\alpha \end{aligned}$$

$$\begin{aligned} \cos 2\alpha &= \cos^2 \alpha - \sin^2 \alpha = 1 - 2 \sin^2 \alpha \\ \sin^2 \alpha &= \frac{1 - \cos 2\alpha}{2} \end{aligned}$$

$$\begin{aligned} &\stackrel{*}{=} -4 \int \frac{1}{2} - \frac{1}{2} \cos 2\alpha d\alpha = \\ &= -4 \left(\frac{\alpha}{2} - \frac{1}{2} \int \cos 2\alpha d\alpha \right) = \\ &= -4 \left(\frac{\alpha}{2} - \frac{1}{4} \sin 2\alpha \right) + C = \\ &= -4 \left(\frac{\alpha}{2} - \frac{1}{4} \cdot 2 \sin \alpha \cos \alpha \right) + C = \\ &= 2 \sin \alpha \cos \alpha - 2\alpha + C = \\ &= 2 \cdot \frac{\sqrt{4-x^2}}{2} \cdot \frac{x}{2} - 2 \arccos \frac{x}{2} + C = \\ &= \frac{x}{2} \cdot \sqrt{4-x^2} - 2 \arccos \frac{x}{2} + C \end{aligned}$$

Exercício 11d

$$\int \sqrt{4+x^2} dx$$



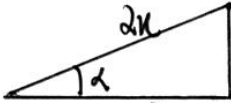
$$\begin{aligned} \int \sqrt{4+x^2} dx &= \int 2 \sec \alpha \cdot 2 \sec^2 \alpha d\alpha = \\ &= 4 \int \sec^3 \alpha d\alpha = \dots = 2 \left(\alpha + \sec \alpha \cdot \operatorname{tg} \alpha \right) + C = \end{aligned}$$

$$\begin{aligned} \alpha &= \arctg \left(\frac{x}{2} \right); \operatorname{tg} \alpha = \frac{x}{2}; \\ \sec \alpha &= \frac{\sqrt{4+x^2}}{2} \end{aligned}$$

$$\begin{aligned} &= 2 \left(\arctg \left(\frac{x}{2} \right) + \frac{\sqrt{4+x^2}}{2} \cdot \frac{x}{2} \right) + C = \\ &= 2 \arctg \left(\frac{x}{2} \right) + \frac{x \sqrt{4+x^2}}{2} + C \end{aligned}$$

Exercício 11e

$$\int \frac{dx}{x^2 \sqrt{4x^2 - 9}}$$



$$\sec \alpha = \frac{2x}{3}$$

$$\tan \alpha = \frac{\sqrt{4x^2 - 9}}{3}$$

$$x = \frac{3}{2} \sec \alpha$$

$$\alpha = \arccos\left(\frac{2x}{3}\right)$$

$$dx = \frac{3}{2} \sec \alpha \tan \alpha d\alpha$$

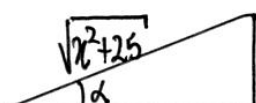
$$\int \frac{dx}{x^2 \sqrt{4x^2 - 9}} = \int \frac{\frac{3}{2} \sec \alpha \tan \alpha d\alpha}{\frac{9}{4} \sec^2 \alpha \cdot 3 \tan \alpha} =$$

$$= \int \frac{3}{2} \cdot \frac{4}{9} \cdot \frac{1}{3} \cos \alpha d\alpha = 2 \cdot \sin \alpha + C =$$

$$= 2 \cdot \frac{\sqrt{4x^2 - 9}}{2x} + C = \frac{\sqrt{4x^2 - 9}}{x} + C.$$

Exercício 11f

$$\int \frac{dx}{x^2 \sqrt{x^2 + 25}}$$



$$\tan \alpha = \frac{x}{5}$$

$$\sec \alpha = \frac{\sqrt{x^2 + 25}}{5}$$

$$x = 5 \tan \alpha$$

$$dx = 5 \sec^2 \alpha d\alpha$$

$$\int \frac{5 \sec^2 \alpha d\alpha}{25 \tan \alpha \cdot 5 \sec \alpha} = \frac{1}{25} \int \frac{\sec \alpha d\alpha}{\tan \alpha} =$$

$$= \frac{1}{25} \int \frac{1}{\cos \alpha} \cdot \frac{\cos^2 \alpha}{\sin^2 \alpha} d\alpha =$$

$$= \frac{1}{25} \int \frac{\cos \alpha}{\sin^2 \alpha} d\alpha =$$

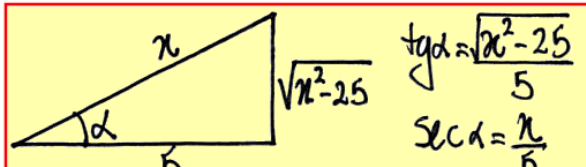
$$u = \sin \alpha \Rightarrow du = \cos \alpha d\alpha$$

$$= \frac{1}{25} \int \frac{du}{u^2} = \frac{1}{25} \cdot \frac{u^{-1}}{-1} + C =$$

$$= -\frac{1}{25} \cdot \frac{1}{\sin \alpha} + C = -\frac{1}{25} \cdot \frac{\sqrt{x^2 + 25}}{x} + C.$$

Exercício 11g

$$\int \frac{x^2}{\sqrt{x^2-25}} dx =$$



$\tan \alpha = \frac{\sqrt{x^2-25}}{5}$
 $\sec \alpha = \frac{x}{5}$
 $x = 5 \sec \alpha \Rightarrow \frac{dx}{d\alpha} = 5 \sec \alpha \cdot \tan \alpha$
 $dx = 5 \sec \alpha \cdot \tan \alpha d\alpha$

$$= \int \frac{(5 \sec \alpha)^2}{5 \tan \alpha} (5 \sec \alpha \cdot \tan \alpha d\alpha) =$$

$$= 25 \int \sec^3 \alpha d\alpha =$$

De exercício anterior:

$$\int \sec^3 x dx = \frac{1}{2} (\sec x \cdot \tan x + \ln |\sec x + \tan x|) + C$$

$$= \frac{25}{2} (\sec \alpha \cdot \tan \alpha + \ln |\sec \alpha + \tan \alpha|) + C =$$

$$= \frac{25}{2} \left(\frac{x}{5} \cdot \frac{\sqrt{x^2-25}}{5} + \ln \left| \frac{x}{5} + \frac{\sqrt{x^2-25}}{5} \right| \right) + C$$

Integral por frações parciais

Exercício 14a

$$\int \frac{2x-1}{x^2-1} dx$$

$$\begin{aligned} \frac{2x-1}{(x+1)(x-1)} &= \frac{A}{x+1} + \frac{B}{x-1} = \frac{Ax-A+Bx+B}{(x+1)(x-1)} = \\ &= \frac{x(A+B)+(-A+B)}{(x+1)(x-1)} \quad \begin{cases} A+B=2 \\ -A+B=-1 \end{cases} \\ &\quad 2B=1 \Rightarrow B=\frac{1}{2}, A=\frac{3}{2} \\ \frac{2x-1}{(x+1)(x-1)} &= \frac{3/2}{x+1} + \frac{1/2}{x-1} \end{aligned}$$

$$\begin{aligned} \int \frac{2x-1}{x^2-1} dx &= \int \frac{3/2}{x+1} dx + \int \frac{1/2}{x-1} dx = \\ &= \left[\frac{3}{2} \cdot \ln|x+1| + \frac{1}{2} \cdot \ln|x-1| + C \right] \end{aligned}$$

Exercício 14b

$$\int \frac{3x-1}{(x-2)(x+5)} dx$$

$$\begin{aligned} \frac{3x-1}{(x-2)(x+5)} &= \frac{A}{x-2} + \frac{B}{x+5} = \frac{Ax+5A+Bx-2B}{(x-2)(x+5)} = \\ &= \frac{Ax+5A+Bx-2B}{(x-2)(x+5)} = \frac{x(A+B)+(5A-2B)}{(x-2)(x+5)} \\ \begin{cases} A+B=3 \\ 5A-2B=-1 \end{cases} &\quad \begin{cases} B=3-A \\ 5A-2(3-A)=-1 \end{cases} \Rightarrow A=\frac{5}{7} \\ &\quad B=3-\frac{5}{7}=\frac{16}{7} \\ \frac{3x-1}{(x-2)(x+5)} &= \frac{5/7}{x-2} + \frac{16/7}{x+5} \end{aligned}$$

$$\int \frac{3x-1}{(x-2)(x+5)} dx = \frac{5}{7} \cdot \ln|x-2| + \frac{16}{7} \cdot \ln|x+5| + C$$

Exercício 14c

$$\int \frac{2x+4}{x^2(x-2)} dx$$

$$\begin{aligned} \frac{2x+4}{x^2(x-2)} &= \frac{A}{x} + \frac{B}{x^2} + \frac{C}{x-2} = \\ &= \frac{Ax(x-2)+B(x-2)+Cx^2}{x^2(x-2)} = \\ &= \frac{Ax^2-2Ax+Bx-2B+Cx^2}{x^2(x-2)} = \\ &= \frac{x^2(A+C)+x(-2A+B)-2B}{x^2(x-2)} \\ \begin{cases} A+C=0 \\ -2A+B=2 \\ -2B=4 \end{cases} &\quad \begin{cases} B=-2 \\ A=-2 \\ C=2 \end{cases} \\ \frac{2x+4}{x^2(x-2)} &= \frac{-2}{x} + \frac{(-2)}{x^2} + \frac{2}{x-2} \end{aligned}$$

$$\begin{aligned} \int \frac{2x+4}{x^2(x-2)} dx &= \int \frac{-2}{x} dx + \int \frac{-2}{x^2} dx + \int \frac{2}{x-2} dx = \\ &= -2 \ln|x| - 2 \cdot \frac{x^{-1}}{-1} + 2 \ln|x-2| + C \end{aligned}$$

Exercício 14d

$$\int \frac{5}{x(x+3)(x-3)} dx$$

$$\begin{aligned} \frac{5}{x(x+3)(x-3)} &= \frac{A}{x} + \frac{B}{x+3} + \frac{C}{x-3} = \\ &= \frac{A(x^2-9) + Bx(x-3) + Cx(x+3)}{x(x+3)(x-3)} = \\ &= \frac{Ax^2 - 9A + Bx^2 - 3Bx + Cx^2 + 3Cx}{x(x+3)(x-3)} = \\ &= \frac{x^2(A+B+C) + x(-3B+3C) - 9A}{x(x+3)(x-3)} \end{aligned}$$

$$\begin{cases} A+B+C=0 & A=-\frac{5}{9} \\ -3B+3C=0 & B=C \\ -9A=5 & -\frac{5}{9} + 2B=0 \Rightarrow B=\frac{5}{18} \\ & C=\frac{5}{18} \end{cases}$$

$$\frac{5}{x(x+3)(x-3)} = \frac{-\frac{5}{9}}{x} + \frac{\frac{5}{18}}{x+3} + \frac{\frac{5}{18}}{x-3}$$

$$\int \frac{5}{x(x+3)(x-3)} dx = \frac{-5}{9} \ln|x| + \frac{5}{18} \ln|x+3| + \frac{5}{18} \ln|x-3| + C.$$

Exercício 14e

$$\int \frac{dx}{x(x^2+1)}$$

$$\begin{aligned} \frac{1}{x(x^2+1)} &= \frac{A}{x} + \frac{Bx+C}{x^2+1} = \frac{Ax^2+A+Bx^2+Cx}{x(x^2+1)} = \\ &= \frac{x^2(A+B) + Cx + A}{x(x^2+1)} \end{aligned}$$

$$\begin{cases} A+B=0 \\ C=0 \\ A=1 \end{cases} \quad B=-1$$

$$\frac{1}{x(x^2+1)} = \frac{1}{x} + \frac{(-x)}{x^2+1}$$

$$\int \frac{1}{x(x^2+1)} dx = \int \frac{1}{x} dx + \int \frac{-x}{x^2+1} dx$$

$$\int \frac{x}{x^2+1} dx = \int \frac{du}{2u} = \frac{1}{2} \ln|u| + C =$$

$$u = x^2+1 \quad = \frac{1}{2} \ln(x^2+1) + C.$$

$$\frac{du}{dx} = 2x \Rightarrow \frac{du}{2} = x dx$$

$$\int \frac{1}{x(x^2+1)} dx = \ln|x| - \frac{1}{2} \ln(x^2+1) + C.$$

Exercício 14f

$$\frac{x^2+x-2}{x^2(3x-1)+(3x-1)} = \frac{x^2+x-2}{(3x-1)(x^2+1)} =$$

$$= \frac{A}{3x-1} + \frac{Bx+C}{x^2+1} = \frac{A(x^2+1)+(Bx+C)(3x-1)}{(3x-1)(x^2+1)} =$$

$$= \frac{Ax^2+A+3Bx^2-Bx+3Cx-C}{(3x-1)(x^2+1)} =$$

$$= \frac{x^2(A+3B)+x(-B+3C)+(A-C)}{(3x-1)(x^2+1)}$$

$$\begin{cases} A+3B=1 & A=C-2 \\ -B+3C=1 & C-2+3B=1 \Rightarrow C=-3B+3 \\ A-C=-2 & -B+3(-3B+3)=1 \\ & -10B=-8 \Rightarrow B=\frac{4}{5} \\ & C=-3 \cdot \frac{4}{5} + 3 = \frac{3}{5} \\ & A=\frac{3}{5}-2 = -\frac{7}{5} \end{cases}$$

$$\frac{x^2+x-2}{(3x-1)(x^2+1)} = \frac{-\frac{7}{5}}{3x-1} + \frac{\frac{4}{5}x+\frac{3}{5}}{x^2+1}$$

$$\int \frac{x^2+x-2}{(3x-1)(x^2+1)} dx = \frac{-7}{5} \int \frac{1}{3x-1} dx +$$

$$+ \frac{1}{5} \int \frac{x+3}{x^2+1} dx$$

$$\int \frac{1}{3x-1} dx \quad u=3x-1 \quad \frac{du}{dx}=3$$

$$dx = \frac{du}{3}$$

$$\int \frac{1}{u} \frac{du}{3} = \frac{1}{3} \ln|u| + C = \frac{1}{3} \ln|3x-1| + C.$$

$$\int \frac{x+3}{x^2+1} dx = \int \frac{x}{x^2+1} dx + \int \frac{3}{x^2+1} dx$$

$$\int \frac{x}{x^2+1} dx = \begin{matrix} u=x^2+1 \\ du=2x dx \\ \frac{du}{2}=x dx \end{matrix}$$

$$= \int \frac{\frac{du}{2}}{u} = \frac{1}{2} \ln|u| + C = \frac{1}{2} \ln(x^2+1) + C$$

$$\int \frac{x^2+x-2}{(3x-1)(x^2+1)} dx = \frac{-7}{15} \ln|3x-1| + \frac{1}{10} \ln(x^2+1) + 3 \arctan x + C.$$

Exercício 14g parte 1

$$\int \frac{\cos x \, dx}{\sin x \sqrt{1+\sin^2 x}} =$$

O fator mais complexo do integrando é o termo $\sqrt{1+\sin^2 x}$, de forma que é interessante alterar a variável para $u=1+\sin^2 x$

$$\frac{du}{dx} = 2 \sin x \cos x \Rightarrow \frac{du}{2} = \sin x \cos x \, dx$$

Para permitir a troca precisamos multiplicar o numerador e o denominador por $\sin x$:

$$= \int \frac{\sin x \cos x \, dx}{\sin^2 x \sqrt{1+\sin^2 x}} = \boxed{\sin^2 x = u-1}$$

$$= \int \frac{du/2}{(u-1)\sqrt{u}} =$$

Para resolver esta integral precisamos de outra substituição:

$$v = \sqrt{u} = u^{1/2} \Rightarrow \frac{dv}{du} = \frac{1}{2} \cdot u^{-1/2}$$

$$2 \, dv = \frac{1}{\sqrt{u}} \, du \quad v = \sqrt{u} \Rightarrow v^2 = u$$

$$= \int \frac{1}{2} \cdot \frac{1}{(u-1)} \left(\frac{du}{\sqrt{u}} \right) = \frac{1}{2} \int \frac{1}{v^2-1} \cdot (2 \, dv) =$$

$$= \int \frac{1}{(v+1)(v-1)} \, dv =$$

Usando frações parciais:

$$\frac{1}{(v-1)(v+1)} = \frac{A}{v+1} + \frac{B}{v-1} = \frac{A(v-1)+B(v+1)}{(v+1)(v-1)} =$$

$$= \frac{v(A+B)-A+B}{(v+1)(v-1)} \quad \begin{cases} A+B=0 \therefore A=-B \\ -A+B=1 \end{cases}$$

$$2B=1 \Rightarrow B=\frac{1}{2} \text{ e } A=-\frac{1}{2}$$

Exercício 14g parte 2

$$\begin{aligned}
 &= \int \frac{-\frac{1}{2}}{v+1} dv + \int \frac{\frac{1}{2}}{v-1} dv = -\frac{1}{2} \cdot \ln|v+1| + \\
 &+ \frac{1}{2} \cdot \ln|v-1| + C = \frac{1}{2} \cdot \ln \left| \frac{v-1}{v+1} \right| + C = \\
 &= \frac{1}{2} \cdot \ln \left| \frac{\sqrt{u}-1}{\sqrt{u}+1} \right| + C = \frac{1}{2} \ln \left| \frac{\sqrt{1+\sin^2 x}-1}{\sqrt{1+\sin^2 x}+1} \right| + C
 \end{aligned}$$

Exercício 17b

$$\int \frac{x}{x^2-4x+8} dx$$

Completamento do quadrado:
 $x^2-4x+8 = x^2-4x+4+4 = (x-2)^2+4$

$$\int \frac{x}{x^2-4x+8} dx = \int \frac{x}{(x-2)^2+4} dx =$$

$$\begin{aligned}
 &\boxed{u=x-2} \quad \boxed{du=dx} \quad = \int \frac{u+2}{u^2+4} du = \\
 &= \int \frac{u}{u^2+4} du + \int \frac{2}{u^2+4} du = \\
 &\int \frac{u}{u^2+4} du = \int \frac{u}{4\left(\frac{u^2}{4}+1\right)} du = \boxed{v=\frac{u}{2}} \quad \boxed{dv=\frac{du}{2}} \\
 &= \int \frac{2v(2dv)}{4(v^2+1)} = \int \frac{v dv}{v^2+1} = \\
 &\boxed{w=v^2+1} \Rightarrow dw=2v dv \Rightarrow \frac{dw}{2}=v dv \\
 &= \int \frac{dw/2}{w} = \frac{1}{2} \ln|w| + C = \frac{1}{2} \ln(v^2+1) + C = \\
 &= \frac{1}{2} \ln\left(\frac{u^2}{4}+1\right) + C = \\
 &= \frac{1}{2} \ln(u^2+4) + C. \\
 &= \frac{1}{2} \ln(u^2+4) + 2 \operatorname{arctg} u + C =
 \end{aligned}$$

$$= \frac{1}{2} \ln((x-2)^2+4) + 2 \operatorname{arctg}(x-2) + C$$

Exercício 15

$$\begin{aligned}
 \int \sec x dx &= \int \frac{1}{\cos x} dx = \int \frac{\cos x}{\cos^2 x} dx = \\
 &= \int \frac{\cos x}{1-\sin^2 x} dx \neq
 \end{aligned}$$

$$\boxed{u=\sin x \Rightarrow \frac{du}{dx} = \cos x \Rightarrow du = \cos x dx}$$

$$\begin{aligned}
 &\neq \int \frac{du}{1-u^2} = \int \frac{du}{(1+u)(1-u)} \stackrel{\text{Fração}}{=} \text{Parciais} \\
 &= \int \frac{1}{1+u} - \frac{1}{1-u} du = \ln|1+u| - \ln|1-u| + \\
 &+ C = \ln \left| \frac{1+u}{1-u} \right| + C = \ln \left| \frac{1+\sin x}{1-\sin x} \right| + C.
 \end{aligned}$$

Exercício 17c

$$\int \frac{1}{\sqrt{5-4x-2x^2}} dx =$$

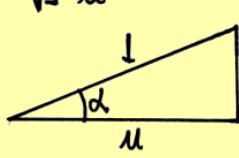
$$\begin{aligned} -2x^2-4x+5 &= -2\left(x^2+2x-\frac{5}{2}\right) = \\ &= -2\left(x^2+2x+1-\frac{7}{2}\right) = -2\left((x+1)^2-\frac{7}{2}\right) = \\ &= 7-2(x+1)^2 \end{aligned}$$

$$= \int \frac{1}{\sqrt{7-2(x+1)^2}} dx =$$

$$\begin{aligned} u &= \sqrt{\frac{2}{7}} \cdot (x+1) \\ du &= \sqrt{\frac{2}{7}} dx \\ dx &= \sqrt{\frac{7}{2}} du \end{aligned}$$

$$\begin{aligned} &= \int \frac{1}{\sqrt{7}(\sqrt{1-u^2})} \cdot \sqrt{\frac{7}{2}} du = \\ &= \frac{\sqrt{2}}{2} \int \frac{1}{\sqrt{1-u^2}} du = \end{aligned}$$

$$\int \frac{1}{\sqrt{1-u^2}} du =$$



$\sin \alpha = \sqrt{1-u^2}$
 $\cos \alpha = u$
 $du = -\sin \alpha d\alpha$

$$= \int \frac{-\sin \alpha d\alpha}{\sin \alpha} = -\alpha + C = -\arccos u + C$$

$$= \frac{\sqrt{2}}{2} (-\arccos u) + C =$$

$$= -\frac{\sqrt{2}}{2} \cdot \arccos\left(\sqrt{\frac{2}{7}}(x+1)\right) + C.$$

Exercício 17h

$$\int \frac{x^2}{x^2-3x+2} dx =$$

$$\begin{array}{r} x^2 \quad | \quad x^2-3x+2 \\ -x^2+3x-2 \quad | \\ \hline 3x-2 \end{array}$$

$$x^2 = (x^2-3x+2) + 3x-2$$

$$= \int \frac{(x^2-3x+2) + (3x-2)}{x^2-3x+2} dx =$$

$$= \int 1 + \frac{3x-2}{x^2-3x+2} dx = x + \int \frac{3x-2}{x^2-3x+2} dx =$$

$$\begin{aligned} x^2-3x+2 &= 0 \quad \Delta = 9-4 \cdot 1(2) = 1 \\ x &= \frac{3 \pm 1}{2} \rightarrow x_1 = 2 \\ &\quad \quad \quad \rightarrow x_2 = 1 \\ x^2-3x+2 &= (x-2)(x-1) \end{aligned}$$

$$\begin{aligned} \int \frac{3x-2}{x^2-3x+2} dx &= \int \frac{3x-2}{(x-2)(x-1)} dx \\ \frac{3x-2}{(x-2)(x-1)} &= \frac{A}{x-2} + \frac{B}{x-1} = \frac{Ax-A+Bx-2B}{(x-2)(x-1)} = \\ &= \frac{x(A+B)-A-2B}{(x-2)(x-1)} \quad \begin{cases} A+B=3 \\ -A-2B=-2 \end{cases} \end{aligned}$$

$$\begin{aligned} A &= 2-2B \\ 2-2B+B &= 3 \Rightarrow B = -1 \\ A &= 2+2 = 4 \end{aligned}$$

$$\frac{3x-2}{(x-2)(x-1)} = \frac{4}{x-2} + \frac{-1}{x-1}$$

$$\begin{aligned} \int \frac{3x-2}{(x-2)(x-1)} dx &= \int \frac{4}{x-2} dx - \int \frac{1}{x-1} dx = \\ &= 4 \ln|x-2| - \ln|x-1| + C. \end{aligned}$$

$$\int \frac{x^2}{x^2-3x+2} dx = x + 4 \ln|x-2| - \ln|x-1| + C.$$

Integral por Completamento de Quadrado

Exercício 18a

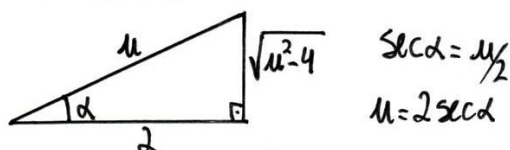
$$\int \frac{dx}{\sqrt{x^2+4x}}$$

$$x^2+4x+4-4=0$$

$$(x+2)^2-4$$

Para usar substituição Trigonométrica é necessário antes completar o quadrado.

Utilizando a substituição $u=x+2$, temos $du=dx$.



$$\tan \alpha = \frac{\sqrt{u^2-4}}{2}$$

$$\frac{du}{d\alpha} = 2 \sec \alpha \cdot \tan \alpha$$

$$\sqrt{u^2-4} = 2 \cdot \tan \alpha$$

$$du = 2 \sec \alpha \tan \alpha d\alpha$$

$$\int \frac{du}{\sqrt{u^2-4}} = \int \frac{2 \sec \alpha \cdot \tan \alpha}{2 \tan \alpha} d\alpha = \int \sec \alpha d\alpha$$

$$= \dots = \ln |\sec \alpha + \tan \alpha| + C = \ln \left| \frac{u}{2} + \frac{\sqrt{u^2-4}}{2} \right| + C$$

$$+ C = \ln \left| \frac{1}{2} (x+2 + \sqrt{x^2+4x}) \right| + C =$$

$$= \ln |x+2 + \sqrt{x^2+4x}| - \ln 2 + C =$$

$$= \ln |x+2 + \sqrt{x^2+4x}| + C.$$

Exercício 18b

$$\int \sqrt{x^2+4x} dx$$

Completando o quadrado:

$$x^2+4x+4-4 = (x+2)^2-4$$

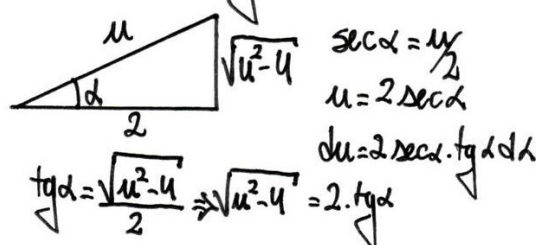
$$\int \sqrt{x^2+4x} dx = \int \sqrt{(x+2)^2-4} dx =$$

$$= \int \sqrt{u^2-4} du$$

$$u = x+2$$

$$du = dx$$

Substituição trigonométrica:



Voltando à integral:

$$\int \sqrt{u^2-4} du = \int (2 \tan \alpha) (2 \sec \alpha \cdot \tan \alpha d\alpha) =$$

$$= 4 \int \sec \alpha \tan^2 \alpha d\alpha = 4 \int \sec \alpha (\sec^2 \alpha - 1) d\alpha =$$

$$= 4 \int \sec^3 \alpha d\alpha - 4 \int \sec \alpha d\alpha *$$

De exercícios anteriores:

$$\int \sec \alpha d\alpha = \ln |\sec \alpha + \tan \alpha| + C$$

$$\int \sec^3 \alpha d\alpha = \frac{1}{2} (\tan \alpha \cdot \sec \alpha + \ln |\sec \alpha + \tan \alpha|) + C$$

$$* 2 \left(\tan \alpha \cdot \sec \alpha - \ln |\sec \alpha + \tan \alpha| \right) + C =$$

$$= 2 \left(\frac{\sqrt{u^2-4}}{2} \cdot \frac{u}{2} - \ln \left| \frac{u}{2} + \frac{\sqrt{u^2-4}}{2} \right| \right) + C =$$

$$= 2 \left(\frac{\sqrt{x^2+4x}}{2} \cdot \frac{(x+2)}{2} - \ln \left| \frac{x+2}{2} + \frac{\sqrt{x^2+4x}}{2} \right| \right) + C =$$

$$= \frac{(x+2)\sqrt{x^2+4x}}{2} - \ln |x+2 + \sqrt{x^2+4x}| + C$$

Exercício 17i

$$\begin{aligned}
 \int \frac{dx}{3x^2 - 2x + 5} &= \int \frac{dx}{3\left(x^2 - \frac{2}{3}x + \frac{5}{3}\right)} = \\
 &= \frac{1}{3} \int \frac{dx}{x^2 - \frac{2}{3}x + \frac{4}{9} + \frac{5}{3} - \frac{4}{9}} = \frac{1}{3} \int \frac{dx}{\left(x - \frac{2}{3}\right)^2 + \frac{11}{9}} = \\
 u &= x - \frac{2}{3} \Rightarrow du = dx \\
 &= \frac{1}{3} \int \frac{du}{u^2 + \frac{11}{9}} = \frac{1}{3} \int \frac{du}{\frac{11}{9}\left(\frac{9}{11}u^2 + 1\right)} = \\
 w &= \frac{3}{\sqrt{11}}u \Rightarrow dw = \frac{3}{\sqrt{11}}du \Rightarrow du = \frac{\sqrt{11}}{3}dw \\
 &= \frac{1}{3} \cdot \frac{9}{11} \int \frac{\frac{\sqrt{11}}{3}dw}{w^2 + 1} = \frac{\sqrt{11}}{11} \int \frac{dw}{w^2 + 1} = \\
 &= \frac{\sqrt{11}}{11} \cdot \arctan w + C = \frac{\sqrt{11}}{11} \cdot \arctan\left(\frac{3}{\sqrt{11}}u\right) + C = \\
 &= \frac{\sqrt{11}}{11} \cdot \arctan\left(\frac{3}{\sqrt{11}}\left(x - \frac{2}{3}\right)\right) + C.
 \end{aligned}$$